

Homework 7

- 1 Consider the Penrose process of a black hole with mass M and angular momentum J . How much energy we can extract from the black hole at most, and what is the most efficient way of extracting energy?
- 2 The Fefferman-Graham expansion for asymptotically AdS spacetime is

$$\frac{ds^2}{\ell^2} = \frac{dr^2}{r^2} + r^2(g_{ij}^{(0)} + r^{-2}g_{ij}^{(2)} + r^{-4}g_{ij}^{(4)} + \dots)dx^i dx^j \quad (1)$$

- 2.1 The outward pointing unit normal vector to the boundary is given by $n_\mu dx^\mu = \frac{dr}{r}$. Write down the boundary stress tensor in terms of $g_{ij}^{(0)}$ and $g_{ij}^{(2)}$.
- 2.2 Let us choose $g_{ij}^{(0)} dx^i dx^j = dx^+ dx^-$, where $x^\pm = \phi \pm t$, the conservation law then allows us to express $g_{ij}^{(2)}$ by two arbitrary functions. Use Einstein equation to check that $g^{(4)}$ is determined by $g^{(2)}$.
- 2.3 The Brown-Henneaux boundary conditions in this gauge is to fix $g_{ij}^{(0)} dx^i dx^j = dx^+ dx^-$. Please find the asymptotic Killing vectors ξ_n and $\bar{\xi}_n$, and check their algebra under the Lie bracket.
- 2.4 The algebra between the charges under Dirac bracket is the same as the algebra of the asymptotic Killing vectors under the Lie bracket up to a central term,

$$\delta_{\xi_m} \mathcal{Q}_{\xi_n} \equiv \{\mathcal{Q}_{\xi_n}, \mathcal{Q}_{\xi_m}\} = \mathcal{Q}_{[\xi_n, \xi_m]} + \int_{\partial\Sigma} k_{\xi_n} [\mathcal{L}_{\xi_m} g, g] \quad (2)$$

where the last term is the central term. Calculate the central charge of the asymptotic symmetry group for 3d Einstein gravity. *hint: For GR calculations, you can use the mathematica package <https://library.wolfram.com/infocenter/MathSource/4484/>*
The charge is given by

$$\delta_\xi Q = \int dx^\alpha \frac{\sqrt{-g} \epsilon_{\mu\nu\alpha}}{16\pi G} [\xi^\mu (\nabla_\sigma \delta g^{\nu\sigma} - \nabla^\nu \delta g_\sigma^\sigma) + \xi_\sigma \nabla^\nu \delta g^{\nu\sigma} + (\frac{1}{2} \delta g_\sigma^\sigma \nabla^\nu - \delta g^{\sigma\nu} \nabla_\sigma) \xi^\mu] \quad (3)$$

3* We can consider AdS/CFT at a finite cutoff. Let us write

$$ds^2 = d\rho + g_{ij}(x^i, \rho) dx^i dx^j \quad (4)$$

and consider a boundary surface at $\rho = \rho_c$. The Brown-York stress tensor is given by

$$T_{ij} = \frac{1}{4G}(K_{ij} - Kg_{ij} + g_{ij}) \quad (5)$$

Use the tangential-normal components of the Einstein equation ($G_{ir} = 0$) to show that the boundary stress tensor is conserved, namely satisfy $\nabla_i T^i_j = 0$. $G_{rr} = 0$ sets up a relation between the trace of the stress tensor T^i_i , and the determinant of T^i_j , please find the relation. *reference: section 3 of arxiv 1801.02714. The relation between the trace and determinant is valid in 3 dimensional bulk. In higher dimensions, the discussion can be found in <https://arxiv.org/pdf/1807.11401.pdf>*